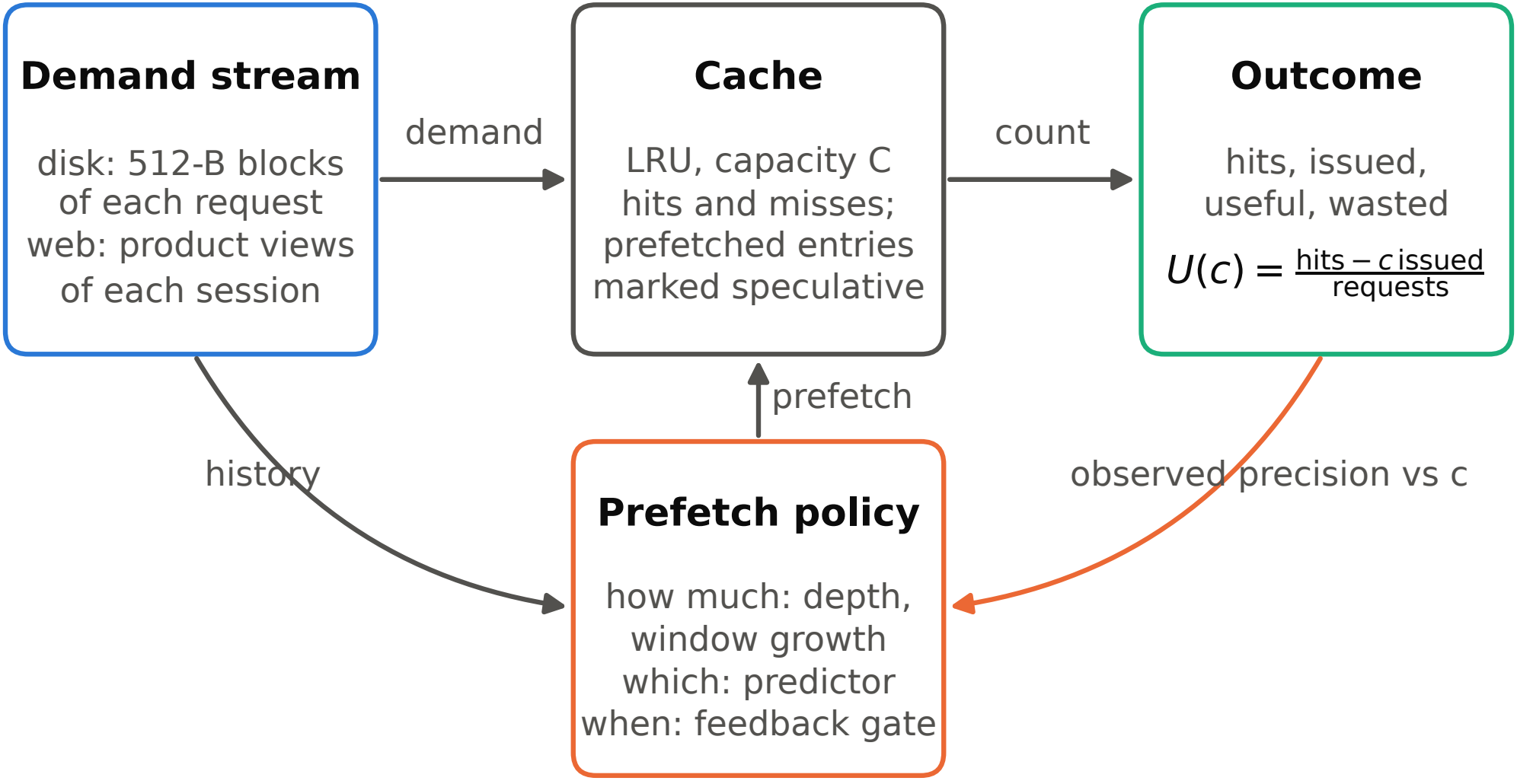
Demand stream
disk: 512-B blocks
of each request
web: product views
of each session
demand
Cache
LRU, capacity C
hits and misses;
prefetched entries
marked speculative
count
Outcome
hits, issued,
useful, wasted
$U(c) = \frac{\text{hits} - c\,\text{issued}}{\text{requests}}$
prefetch
history
observed precision vs c
Prefetch policy
how much: depth,
window growth
which: predictor
when: feedback gate